Physics-Gated Correction Discovery: Why Constraining the Search Space Beats Scaling It in Noisy Symbolic Regression

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Abstract

Symbolic regression (SR) methods such as PySR, AI Feynman, and PhySO discover equations *from scratch*, an approach that overfits on noisy physical data: under matched residuals, PySR's structural accuracy is non-monotonic in noise and does not improve when its search budget is doubled. We argue that scientific discovery is rarely *tabula rasa*—it is *correction-first*: given a known classical law and anomalous data, the real task is to find the minimal physically-valid correction Δ. We present **Anomaly-Driven Correction Discovery (ADCD)**, which restricts search to a correction subspace and cascades three physical filters—AST complexity, dimensional homogeneity with a transcendental-argument guardrail, and asymptotic-regime consistency (ARC)—*before* a JAX optimizer fits any parameter, with Bayesian Information Criterion (BIC) reranking. On a 9-scenario benchmark (overall mean $80.4\% \pm 7.4\%$ across 16 seeds and all noise levels; 88.9% at the disclosed reference seed), ADCD reaches 88.9% at 5% noise versus 11.1% for PySR—a *seed-independent* gap: ADCD's *worst* of 16 seeds still beats PySR with doubled budget. Removing all gates drops recovery by 44.5 pp. ADCD is orthogonal to—not competitive with—existing SR: it restricts the problem they solve to a physically-motivated subspace where they fail under noise.

1 Introduction

When a physical law breaks down under extreme conditions, the scientific question is not "what is the correct law?"—it is "what is the *minimal correction* to the known law?" The dominant paradigm in symbolic regression (SR) answers the wrong question: it discovers $f(X) \approx y$ from a blank slate, an approach that degrades under realistic noise. Pioneered by Eureqa Schmidt & Lipson [2009], AI Feynman Udrescu & Tegmark [2020], and PySR Cranmer [2020], and extended by deep RL approaches such as DSR Petersen et al. [2021] and unit-constrained PhySO Tenachi et al. [2023], these systems recover compact laws in *low*-noise regimes. Under realistic observational noise, however, their structural accuracy degrades and is often *non-monotonic* in noise: a larger candidate pool does not reliably translate into the correct functional *class* (Section 3).

We observe that this difficulty is partly one of *problem formulation*. Science rarely discovers from a blank slate—it *corrects*. From the relativistic kinetic-energy expansion $\frac{1}{2}mv^2(1+\frac{3}{4}v^2/c^2+\dots)$, to Debye-screened Coulomb potentials $V(r) \propto r^{-1}e^{-r/\lambda_D}$, to turbulent drag $F = bv + \alpha v^2$, progress takes a structurally-correct classical law and appends a *minimal correction* Δ that reconciles it with anomalous data. This restricts the hypothesis space from the full function space $\mathcal{F}$ to a correction subspace $\mathcal{D} \subset \mathcal{F}$—an exponential reduction informed by the known physics.

We propose **Anomaly-Driven Correction Discovery (ADCD)**, built on three contributions:

C1 Correction-first paradigm. Search for Δ in $\mathcal{D}$, not f in $\mathcal{F}$, given a classical law $y_{\text{classical}} = g(X;\phi)$ and anomalous observations.

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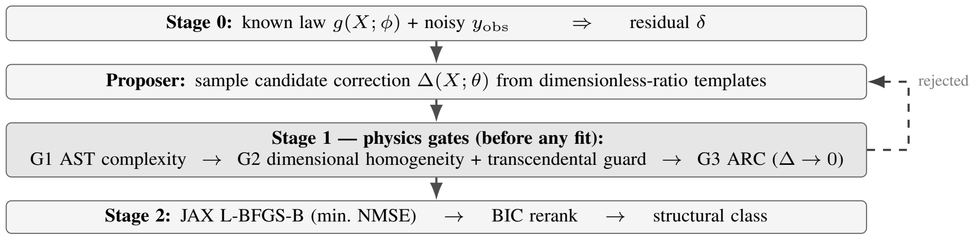


Figure 1: ADCD pipeline. Physics gates (G1–G3) screen candidates *before* any parameter is fit; rejected candidates are re-sampled.

C2 Physics-gated cascade before optimization. Three filters—AST complexity, dimensional homogeneity with a transcendental-argument guardrail, and asymptotic-regime consistency (ARC, $\Delta \to 0$ at the classical boundary)—screen candidates *before* any parameter is fit.

C3 Categorical immunity to trigonometric overfitting. Because dimensional/ARC gates reject nested $\sin(\sin(\dots))$ that dominate unconstrained Pareto fronts at 0% noise, ADCD's accuracy is stable where PySR's is non-monotonic.

We deliberately scope this paper to *method and synthetic benchmark*; real-data validation (galactic, cosmological) is reported in a companion manuscript.

2 The ADCD framework

Residual extraction. Given a known classical law $y_{\text{classical}} = g(X;\phi)$ and noisy anomalous data y_{obs}, ADCD isolates the correction by forming a residual. In *multiplicative* mode $y_{\text{true}} = y_{\text{classical}}(1+\Delta)$ gives $\delta = y_{\text{obs}}/y_{\text{classical}} - 1$; in *additive* mode $y_{\text{true}} = y_{\text{classical}} + \Delta$ gives $\delta = y_{\text{obs}} - y_{\text{classical}}$. A mode selector picks whichever best isolates the anomalous deviation.

Proposer. Candidate correction templates $\Delta(X;\theta)$ are drawn from a stochastic sampler over dimensionless ratios, biased by residual features (monotonicity, curvature, oscillation, exponential decay R^2, symmetry) that upweight plausible families but never gate candidates—only the physics cascade does.

Stage 1: cascaded physics gates. Every candidate passes three filters *before* optimization:

G1. AST complexity. Reject expressions with AST depth > 7 or > 20 tokens—a coarse Occam prior.

G2. Dimensional homogeneity + transcendental guard. Verify Δ is dimensionless; physical variables may enter only inside dimensionless ratios. As a sub-check, $\sin, \cos, \exp, \log, \tanh$ may take only *dimensionless* arguments: $\exp(-r)$ without a scale θ_1 is blocked instantly. Free parameters carry adaptive units (relaxable), documented in the companion paper.

G3. Asymptotic-regime consistency (ARC). Enforce the decoupling limit: as $x \to x_0$ (e.g. $v \to 0$, $r \to \infty$), $\lim_{x\to x_0} \Delta(X;\theta) = 0$, evaluated with SymPy's limit engine.

Survivors are then coarsely ranked by preliminary NMSE on δ, giving an early data-driven signal before the full fit.

Stage 2: JAX-traced optimization + BIC reranking. Surviving expressions are transpiled to JAX and fit by L-BFGS-B (multi-restart, log-uniform init) minimizing NMSE of the residual, $\text{NMSE} = \frac{\sum_i (\Delta(X_i;\theta)-\delta_i)^2}{\sum_i (\delta_i - \bar{\delta})^2}$. Under noise, flexible wrong-class expressions can achieve lower NMSE than the truth, so we rerank by the Bayesian Information Criterion Schwarz [1978], Rissanen [1978], $\text{BIC} = N \ln(\text{NMSE}) + k \ln N$, whose $k \ln N$ penalty selects parsimonious correct-class forms. The winner is classified into one of six families (exponential, trigonometric, logarithmic, power law, rational, polynomial) by AST traversal, with degenerate-exponent detection (a fitted $\hat{\theta}_1 \approx 1$ recasts $\theta_0 x^{\theta_1}$ as polynomial).

Relation to EFT. The cascade mirrors the correction-first philosophy of Effective Field Theory Weinberg [1979]: dimensional analysis of operators (G2), decoupling of heavy modes in the

Table 1: Structural class match rate on the 9-scenario residual benchmark. `generous` doubles PySR's budget on every axis yet does not close the gap.

Method	0%	1%	5%	10%
ADCD (ours, seed=42)	9/9 (100%)	9/9 (100%)	**8/9 (88.9%)**	8/9 (88.9%)
ADCD 16-seed mean	86.8%	81.3%	*77.1%±10.0%*	76.4%
PySR fair	4/9 (44.4%)	5/9 (55.6%)	1/9 (11.1%)	5/9 (55.6%)
PySR generous (2×)	4/9 (44.4%)	4/9 (44.4%)	5/9 (55.6%)	2/9 (22.2%)

low-energy limit (G3/ARC), and power counting (BIC). ADCD has no renormalization-group flow or controlled E/Λ expansion—it *automates* that philosophy for data-driven problems.

3 Experiments

Benchmark. We evaluate on 9 physical-anomaly scenarios in three tiers: *textbook* (relativistic KE, Yukawa gravity, anharmonic spring), *cross-domain* (net radiation T^{-4}, nonlinear drag, screened Coulomb), and *synthetic novel* formulas. Each is run at $0/1/5/10\%$ observational noise. Success = correct structural *class* recovery (e.g. discovering an exponential correction where the truth is exponential). We report a 16-seed distribution; the mean is the primary claim and seed=42 (the highest seed) is disclosed explicitly, never as the headline.

Baselines. We compare against PySR Cranmer [2020] on the *same residual targets* $\delta = y_{\text{obs}} - y_{\text{classical}}$. The primary baseline is the `fair` profile (n_{iter}=100, `maxsize`=30, 60 s/scenario). To rule out a budget artefact we add a `generous` profile that *doubles* every budget axis (n_{iter}=200, `maxsize`=40, 120 s; $\sim 3\times$ wall-clock).

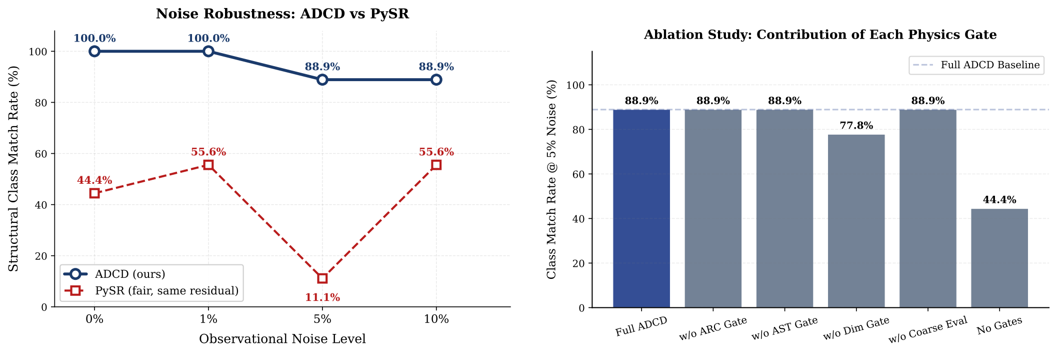
Main result: the gap is structural, not budgetary. Table 1 reports structural match rates. ADCD reaches 88.9% (8/9) at 5% noise versus 11.1% (1/9) for PySR fair—a 77.8 pp gap under matched residuals. The comparison is *seed-independent*: at 5% noise ADCD's worst of 16 seeds (66.7%, 6/9; cf. the per-seed distribution in our bundled artefacts) still exceeds PySR fair. Crucially, doubling PySR's budget does *not* close the gap—`generous` total recovery stays flat at $15/36$ (identical to `fair`) and at the 5% headline point it reaches only $5/9$ (55.6%). The extra budget explores deeper trees (mean complexity 9.8 vs. 8.7 nodes) spent on more deeply-overfit rather than more physically-correct structure. The binding constraint is *structural selection*, not search budget; physics gates resolve it directly.

Why PySR is non-monotonic, and why ADCD is not. PySR fair exhibits a non-monotonic noise profile $(4, 5, 1, 5$ out of 9). At 0% noise the large search space produces deeply-nested trigonometric expressions (mean 13.8 nodes) that fit clean data but are structurally wrong ($5/5$ mismatches are trigonometric); at 1% noise regularization drops complexity to 8.2 nodes; at 5% it picks the wrong family entirely; at 10% it collapses to the simplest form. ADCD avoids this because its gates are noise-independent: dimensional and ARC constraints reject the nested trig that PySR overfits to, and once the correct family is sampled the cascade selects it regardless of SNR (Section 3).

Ablation: the gates act collectively. Section 3 shows the result of disabling gates one at a time at 5% noise. Removing ARC, AST, or the coarse-eval pre-filter individually leaves accuracy unchanged ($8/9$); removing the dimensional gate alone costs 11.1 pp ($8/9 \to 7/9$). The dimensional gate is the most critical single filter because violations of dimensional homogeneity—e.g. $\exp(-r)$ with r in metres—are the most common failure mode in unconstrained SR, while ARC violations are rarer but produce unphysical extrapolation. Only disabling *all* gates causes a larger collapse ($8/9 \to 4/9$, -44.5 pp): the gates reinforce each other as a collective physical-consistency filter rather than each being independently essential.

4 Discussion and limitations

ADCD is a restriction, not a competitor. ADCD does not compete with PhySO or PySR—it restricts the problem they solve to a physically-motivated subspace where they are known to fail



(a) Noise robustness. ADCD stable; PySR fair non-monotonic (44→56→11→56%).

(b) Gate ablation at 5% noise. Removing all gates collapses recovery by 44.5 pp.

Figure 2: Left: structural match rate vs. noise. Right: per-gate ablation.

under noise. This makes ADCD directly *combinable* with them: a PhySO/PySR/LaSR Grayeli et al. [2024] proposer could feed the same gate cascade.

Honest limitations. ADCD assumes a structurally-correct classical baseline (incremental correction, not wholesale replacement). At $\geq 5\%$ noise, screened-Coulomb decay is fitted by a rational proxy, reflecting decay-vs-saturation ambiguity under limited dynamic range. A multivariable extension (Buckingham-Π) is currently weaker ($2/4$ at 5%) and treated as exploratory; the dimensional gate uses adaptive parameter-unit relaxation rather than PhySO's strict enforcement—complementary design points.

Reproducibility. ADCD ships as an open-source Python package (JAX backend, public API/CLI, CI/CD); all 16 seeds, per-seed noise distributions, and PySR profiles are bundled for replication.

Fail-safe behavior. On a 6-case misspecification benchmark, ADCD fails safe—it fails to converge rather than producing confident spurious corrections—a property absent from tabula rasa SR.

Closing. Noisy symbolic regression is bottlenecked by *structural selection*, not search budget. The correction-first formulation—searching for a physically-valid Δ to a known law, screened by dimensional and asymptotic gates before any fit—resolves this directly: ADCD delivers seed-independent gains over PySR, stays robust where unconstrained search is non-monotonic, and is orthogonal to existing SR systems. Future work extends the cascade to multivariable laws and integrates it as a selection layer over external proposers (PhySO, LaSR).

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